Supplementary Material

Architecture and Preprocessing Choices under Dataset Shift in Single-Channel EEG Sleep Staging: A Controlled Evaluation on Sleep-EDF and SHHS1

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This supplement contains implementation details and descriptive tables moved from the main manuscript. The evidence labels are unchanged: seed 42 is primary, seed 123 is a post-protocol sensitivity repeat, SHHS1 E1/E2 analyses are secondary on an opened cohort, and SHHS1 E3–E2/E4 analyses are post-hoc. E6 estimates mean and standard deviation from each complete unlabelled target recording; it is label-free but transductive at recording level, not purely inductive zero-shot inference.

Supplementary Methods S1: model and training details

Architecture and feature construction

Each encoder input is a 30-s epoch with 3,000 samples. The control encoder contains 15 separately trained CNNs: three context groups (current, previous and next; C/P/N) crossed with five class-specialised loss functions. Every CNN has a five-class output, not a binary head. Concatenating its softmax outputs in C/P/N order gives 75 features per central epoch. The P and N inputs are shifted within each recording and are trained against the central epoch's label; the first or last signal epoch is replicated at a recording edge. No context crosses recording boundaries. These are softmax-output features; their use does not imply that probability calibration has been established.

The ResNet-1D encoder instead produces a 128-dimensional, pre-dropout representation of the current epoch. Layer specifications are given in Table S1. Encoders are trained and selected before sequence training. Their selected checkpoints are loaded in evaluation mode, and features are cached without gradients. E1 reuses the E0 encoder and its features within the same fold and seed; its sequence training settings differ from E0 (Table S2). E1–E0 is therefore a sequence-model configuration comparison, not an architecture-only effect. E2–E1 replaces the feature/context package, including encoder training and selection, while retaining the specified TCN recipe.

Table S1: Layer specifications. Convolution entries give input/output channels, kernel, stride and padding; d denotes dilation. All final classifiers have five outputs.

Component	Layers and operations
One control CNN	Conv1d $1 \to 10$, $k = 55, s = 1, p = 27$, no bias; ReLU; batch normalisation; max-pool $k = s = 16$, ceiling mode. Then Conv1d $10 \to 5$, $k = 25, s = 1, p = 12$, no bias; ReLU; batch normalisation; the same max-pool. Flatten $5 \times 12 = 60$; linear $60 \to 10$; ReLU; linear $10 \to 5$.
ResNet stem	Conv1d $1 \to 32$, $k = 50, s = 5, p = 25$, no bias; batch normalisation; ReLU; max-pool $k = 3, s = 2, p = 1$.
ResNet blocks	Three residual blocks: $32 \to 64$ ($s = 1$), $64 \to 128$ ($s = 2$), $128 \to 128$ ($s = 2$). Each has two bias-free $k = 7, p = 3$ convolutions, with the stated stride in the first and stride 1 in the second; batch normalisation after each, ReLU after the first and after residual addition. Each shortcut uses a bias-free $k = 1$ projection with the first convolution's stride, followed by batch normalisation.
ResNet output	Global average pooling gives 128 features. During encoder training only, dropout 0.5 precedes the linear $128 \to 5$ classifier. Cached features are taken before dropout.
BiLSTM	One bidirectional LSTM layer, input 75, hidden size 128 per direction; linear $256 \to 5$ output. Packed sequences exclude batch padding from recurrence.
TCN	Linear input projection 75 or $128 \to 128$; layer normalisation; GELU. Six residual blocks, each with two bias-free $128 \to 128$ convolutions, $k = 3, s = 1, p = d$, with $d \in \{1, 2, 4, 8, 16, 32\}$. Each convolution is followed by per-epoch layer normalisation, GELU and dropout 0.2. The shortcut is identity. Final dropout 0.2 and linear $128 \to 5$.

Training and checkpoint selection

All stages use the same subject partitions, with validation at the end of each epoch, but not the same optimisation or selection rule. For a CNN specialised to class c, training counts n_c and $N = \sum_c n_c$ determine weighted five-class cross-entropy: the weight for c is $0.5/(n_c/N)$ and each other class has weight $0.5/((N - n_c)/N)$. These training-derived weights are also used to evaluate its validation loss. ResNet uses weights $N/(5n_c)$, again computed only from training subjects. Sequence-model cross-entropy is unweighted over valid labelled epochs. CNN branch seeds are the campaign seed plus the branch index in C/P/N–stage order; ResNet and sequence training use the campaign seed. No label smoothing, data augmentation or learning-rate scheduler is used.

Table S2: Specified training recipes. Patience counts end-of-epoch validation checks without an improvement. E1 reuses the E0 encoder; all other reported encoder/sequence pairs are fitted for their own input variant.

Setting	CNN15 (E0/E1)	BiLSTM (E0)	ResNet (E2–E6)	TCN (E1–E6)
Optimiser	Adam	Adam	AdamW	Adam
Learning rate	10^{-3}	10^{-2}	10^{-3}	5×10^{-4}
Weight decay	0	0	10^{-4}	0
Batch size	64 epochs	4 recordings	64 epochs	8 recordings
Maximum epochs	1,000	1,000	40	300
Patience	10	10	8	30
Selected checkpoint	Minimum weighted validation loss	Maximum validation macro-F1	Maximum validation macro-F1	Maximum validation macro-F1
Gradient-norm clipping	None	1.0	None	1.0

Whole retained recordings are the sequence units for training and inference. Batch padding and unknown/movement labels are excluded from the loss and metrics; unknown epochs retain their positions in the signal sequence. The TCN masks padded positions after the input projection and each convolutional sub-block. Its theoretical receptive field is $1 + 2(3 - 1)(1 + 2 + 4 + 8 + 16 + 32) = 253$ feature epochs, spanning up to 126 epochs on each side, subject to recording boundaries. This is not a causal architecture. E1 also receives the explicit neighbouring signal epochs through its CNN features. The 100-epoch input used for timing below is a benchmark shape, not a training-window length.

Signal operations and evaluation windows

Signal values are read in physical microvolts. Sleep-EDF uses Fpz–Cz at 100 Hz. SHHS1 uses C4–A1 and is resampled continuously from 125 to 100 Hz with polyphase resampling (up/down = 4/5, Kaiser window $\beta = 5$, constant padding). Table S3 specifies operations after this cohort-specific loading/resampling step. Filtering uses a fourth-order Butterworth design with 0.5–30 Hz cutoffs, implemented as second-order sections with forward–backward filtering. It operates on the continuous recording before epoch extraction and trimming.

Table S3: Input variants and fitted components. Operations are listed in execution order; all model weights are learned on Sleep-EDF only.

ID	Signal operations	Encoder and sequence configuration
E0	Raw microvolt signal	CNN15 C/P/N, then BiLSTM
E1	Raw microvolt signal	Reused E0 CNN15 C/P/N, then TCN
E2	Raw microvolt signal	Current-epoch ResNet, then TCN
E3	Band-pass; clip at $\pm 800\,\mu$V; divide by 100	Current-epoch ResNet, then TCN
E4	Band-pass	Current-epoch ResNet, then TCN
E5	Band-pass; clip at $\pm 800\,\mu$V	Input-identity audit; excluded from performance analysis
E6	Band-pass; clip at $\pm 800\,\mu$V; recording-wise z-score	Current-epoch ResNet, then TCN

For E6, mean and population standard deviation (ddof = 0) are calculated over the entire filtered and clipped recording, including portions outside the later evaluation window. The implementation rejects zero or non-finite standard deviation rather than adding an epsilon; non-finite input/output signals are also rejected. Processing uses float64 for filtering and normalisation and stores model inputs as float32. On SHHS1 this is label-free target-record normalisation with a transductive dependency; it does not update model weights. E3 and E6 are trained separately on their respective source input variants, so their comparison is not an amplitude-only intervention on one frozen classifier.

The evaluation window extends up to 30 minutes before the first and after the last reference sleep epoch (N1, N2, N3 or REM), clipped to recording limits. Original stage 3/4 labels are merged as N3; unknown/movement epochs are unscored. This reference-dependent trimming convention and the later boundary diagnostics must be distinguished from label-free model fitting/inference. Sleep-EDF metrics use out-of-fold predictions, with each subject evaluated by a model not trained on that subject. SHHS1 uses the arithmetic mean of class probabilities from all ten source-fold models before argmax. Cross-cohort score differences therefore compare distinct cohorts and evaluation settings, not an isolated effect of dataset shift.

Runtime measurement

The archived forward-pass benchmark used the selected fold-00, seed-42 checkpoints on a Tesla V100-PCIE-16GB GPU, Python 3.10.13, PyTorch 2.5.1+cu124, CUDA 12.4 and cuDNN 9.1. It used float32 synthetic input of shape $(1, 100, 1, 3000)$, 20 warm-up passes and 100 timed repetitions in each of three rounds, with model order varied across rounds and GPU synchronisation around timing. Both encoder and sequence forward passes are included; disk I/O, preprocessing, feature-cache serialization and training are excluded. Latency is the median of the recorded passes; memory is the maximum peak allocated GPU memory, distinct from reserved or incremental memory. These measurements characterise offline computation, not the time until future EEG needed by a bidirectional model becomes available. Training-and-validation wall-clock times additionally depend on the recipes and stopping rules in Table S2.

Supplementary results

Table S4: Per-class Sleep-EDF metrics for E3 and E6 (seed 42). Differences are calculated before rounding.

Stage	E3 Prec.	E3 Rec.	E3 F1	E6 Prec.	E6 Rec.	E6 F1	ΔF1
W	.9338	.9271	.9304	.9391	.9169	.9279	+.0026
N1	.5149	.5410	.5276	.5032	.5219	.5124	+.0152
N2	.8583	.8447	.8514	.8401	.8368	.8385	+.0130
N3	.8072	.8097	.8085	.7160	.7779	.7457	+.0628
REM	.8274	.8413	.8343	.8227	.8197	.8212	+.0131

Table S5: Pooled E3 confusion matrix on Sleep-EDF (rows: reference; columns: prediction).

	W	N1	N2	N3	REM
W	61,133	3,888	416	56	448
N1	3,283	11,643	4,867	105	1,624
N2	539	5,496	58,394	2,271	2,432
N3	30	45	2,377	10,558	29
REM	483	1,542	1,984	90	21,736

Table S6: Sleep-EDF pooled macro-F1 under two fixed seeds. Seed 42 is primary; seed 123 is a post-protocol sensitivity repeat. Differences are calculated before rounding.

Configuration	Seed 42	Seed 123	Difference
E0	.7754	.7755	+.0000
E1	.7802	.7776	-.0026
E2	.7835	.7828	-.0007
E3	.7904	.7883	-.0022
E4	.7891	.7887	-.0004
E6	.7691	.7780	+.0089

Table S7: Seed sensitivity of the protocol-specified Sleep-EDF contrasts. Bootstrap intervals describe differences in pooled macro-F1; Wilcoxon tests use paired subject-wise macro-F1. Holm correction is applied to the four tests within each seed, not to the marginal bootstrap intervals.

Contrast	Seed 42: Δ [95% CI], p_{Holm}	Seed 123: Δ [95% CI], p_{Holm}
E1–E0	.004811 [−.000915, .010193], .1022	.002188 [−.002870, .007043], .9130
E2–E1	.003251 [−.002370, .008808], .1036	.005143 [−.000571, .011292], .2400
E3–E2	.006962 [.000305, .014520], .8989	.005478 [−.002611, .015796], .9130
E3–E6	.021319 [.012179, .030698], .0012	.010249 [.002435, .017989], .1313

Table S8: Seed-123 wall-clock training and validation time (10 folds, sequential, same GPU). E1 time covers sequence training only because its encoder and cached features are reused from E0.

Configuration	Total	Mean per fold
E0: 15-CNN + BiLSTM	33 h 35 min 36 s	3 h 21 min 34 s
E1: 15-CNN + TCN	14 min 17 s	1 min 26 s
E2: ResNet-1D + TCN	2 h 44 min 57 s	16 min 30 s
E3: fixed scale	3 h 16 min 40 s	19 min 40 s
E4: band-pass	2 h 55 min 56 s	17 min 36 s
E6: record-wise z-score	2 h 24 min 44 s	14 min 28 s

Table S9: Seed-123 sensitivity and E4 extension on the same 180 SHHS1 participants. These are post-protocol results on the previously evaluated cohort.

ID	Subject-mean macro-F1	Pooled macro-F1	Accuracy	κ
E0	.5303	.5785	.6738	.5420
E2	.5409	.5858	.6874	.5597
E3	.5634	.6031	.7003	.5772
E4	.5732	.6147	.7064	.5869
E6	.5503	.5865	.6873	.5552

Table S10: Paired SHHS1 seed-123 extension. Differences and subject-cluster percentile bootstrap intervals concern subject-mean macro-F1 (10,000 resamples). Two-sided Wilcoxon p-values are Holm-adjusted within this four-contrast extension, separately from the primary seed-42 family. W/T/L denotes subjects with a positive/zero/negative paired difference.

Contrast	Δ subject-mean macro-F1	95% CI	p_{Holm}	W/T/L
E4–E2	+.0323	[.0236, .0410]	7.40×10^{-13}	135/1/44
E3–E4	-.0098	[−.0125, −.0073]	1.89×10^{-10}	56/1/123
E3–E6	+.0131	[.0023, .0239]	.0085	111/0/69
E3–E0	+.0331	[.0225, .0438]	5.19×10^{-9}	131/0/49

Table S11: Secondary and post-hoc SHHS1 paired contrasts (seed 42; 180 subjects). Intervals are marginal 95% paired subject-cluster percentile bootstrap intervals for the subject-mean difference, with 10,000 resamples. The E1/E2 extension was specified before their SHHS1 inference but after the cohort had been evaluated with E0/E3/E6. E3–E2 was specified after their separate marginal results had been examined.

Contrast	Δ subject-mean macro-F1	95% CI	p	W/T/L
E1–E0	+.00651	[.00192, .01087]	.00325[a]	108/0/72
E2–E1	-.01280	[−.02209, −.00350]	.01005[a]	80/0/100
E3–E2	+.04750	[.03724, .05792]	2.35×10^{-17} [b]	147/0/33

[a]Two-sided subject-wise Wilcoxon, Holm-adjusted within the two-contrast E1/E2 extension (bootstrap seed 2031). [b]Unadjusted two-sided Wilcoxon for one separately specified post-hoc comparison (bootstrap seed 2032); it is not part of either historical Holm family. The contrasts compare specified pipeline configurations, not isolated architectural effects.

Table S12: Per-class transfer of E3; Pred/True is the predicted-to-reference class count ratio, not a measure of probability calibration. Differences are calculated before rounding.

	Sleep-EDF		SHHS1				
Stage	F1	Pred/True	Prec.	Rec.	F1	Pred/True	ΔF1
W	.9304	.993	.8823	.8014	.8399	.908	-.0905
N1	.5276	1.051	.2528	.5437	.3451	2.151	-.1824
N2	.8514	.984	.7394	.7349	.7372	.994	-.1143
N3	.8085	1.003	.9440	.2582	.4055	.273	-.4030
REM	.8343	1.017	.6052	.8939	.7218	1.477	-.1125

Table S13: Pooled E3 confusion matrix on SHHS1 (rows: reference; columns: prediction).

	W	N1	N2	N3	REM
W	29,638	4,481	1,171	21	1,672
N1	874	3,807	601	3	1,717
N2	2,348	5,947	56,063	324	11,605
N3	78	39	16,674	5,888	127
REM	655	784	1,311	1	23,183

Table S14: Per-class SHHS1 metrics for E3 and E2 (seed 42). Differences are calculated before rounding.

Stage	E3 Prec.	E3 Rec.	E3 F1	E2 Prec.	E2 Rec.	E2 F1	ΔF1
W	.8823	.8014	.8399	.8780	.7275	.7957	+.0442
N1	.2528	.5437	.3451	.2367	.3528	.2833	+.0618
N2	.7394	.7349	.7372	.7362	.6937	.7143	+.0228
N3	.9440	.2582	.4055	.9659	.2496	.3967	+.0087
REM	.6052	.8939	.7218	.4886	.9450	.6442	+.0776

Table S15: Reference-label oracle on E3 predictions (seed 42). Each row corrects only the indicated confusion-matrix cell. The percentage is relative to the observed pooled-score difference between Sleep-EDF out-of-fold evaluation and SHHS1 ensemble evaluation, not an isolated domain-shift effect or an estimate of achievable model performance.

Condition	macro-F1	Numerical gap recovered
Observed E3 on SHHS1	.6099	—
Correct N3→N2 only	.7444	.1345 (74.5%)
Correct N2→REM only	.6596	.0497 (27.5%)
E3 on Sleep-EDF	.7904	—

Table S16: Post-hoc reference-label boundary and stable-interior diagnostics. Seed 123 reuses the same subjects and label-defined regions; it is a sensitivity repeat, not an independent cohort. All metrics are pooled within each region.

Cohort	Seed	ID	Boundary macro-F1	Stable macro-F1	Boundary N3 recall	Stable N3 recall
Sleep-EDF	42	E0	.6179	.8175	.6121	.9216
Sleep-EDF	42	E3	.6321	.8346	.6326	.9316
Sleep-EDF	123	E0	.6181	.8173	.6174	.9162
Sleep-EDF	123	E3	.6343	.8299	.6385	.9285
SHHS1	42	E0	.4323	.5959	.0633	.4008
SHHS1	42	E3	.4689	.6220	.0733	.3900
SHHS1	123	E0	.4336	.5957	.0639	.3885
SHHS1	123	E3	.4605	.6166	.0656	.3660

For a label change between consecutive valid epochs $j-1$ and j, where j is the first epoch with the new label, the boundary neighbourhood contains $\{j-1, j, j+1\}$ within the same contiguous recording segment. Overlapping neighbourhoods are combined. Unknown-label gaps and recording boundaries break contiguity. Stable-interior epochs are at least three epoch positions from *both* sides of every label change, using distance $\min(|i-(j-1)|, |i-j|)$; segments without a change are stable. The two regions do not exhaust the scored epochs: Sleep-EDF contains 43,237 boundary, 130,693 stable and 21,539 remaining epochs; SHHS1 contains 44,744, 103,195 and 21,073, respectively. The remaining epochs are not simply the distance-two bin, because the boundary neighbourhood is centred on the first new-label epoch. These masks include all stage changes, not just N2–N3. Where persistent boundaries are reported in the archived diagnostics, each side must contain at least three consecutive epochs of its respective label. Region metrics retain the fixed five-class convention used in the main analysis.

E6 N3 recall is 0.2005 overall and 0.0721 in the same SHHS1 boundary neighbourhood (seed 42), compared with 0.2582 and 0.0733 for E3. These regions use reference labels on both sides and are offline diagnostics, not direct measurements of physiological transitions or deployable detectors. Differences in class composition between regions also preclude interpreting their score difference as an isolated effect of proximity to a label change. N3 under-detection persists in SHHS1 stable interiors in both seeds.

Supplementary analyses: representation and context

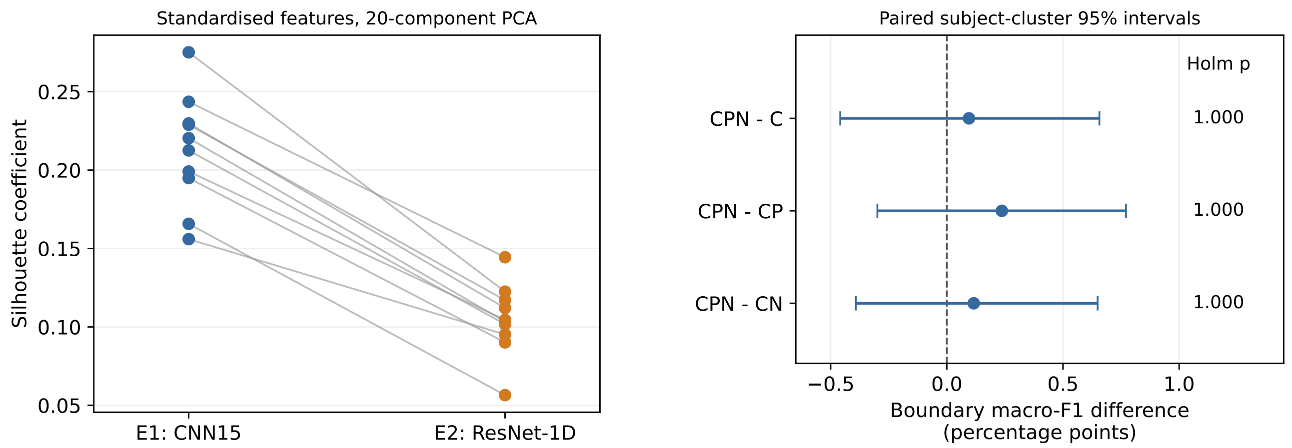
The representation analysis uses the same 1,000 held-out epochs per fold for E1 and E2: 200 per stage, sampled without replacement with a round-robin allocation across available subjects (sampling seed 42 + fold index). In each representation and fold separately, features are standardised and projected onto 20 principal components; these transforms are fitted to the sampled test features solely for this descriptive analysis, not fed back into the staging model. Silhouette uses Euclidean distance in the resulting space and the reference stage labels. The mean coefficient across ten folds is 0.2126 (SD 0.0356) for E1 and 0.1048 (SD 0.0229) for E2. E1 consists of 75 softmax-output features, whereas E2 is an unconstrained 128-dimensional representation; this difference and the analysis transforms prevent treating silhouette as a like-for-like measure of downstream predictive utility.

For the context ablation, the pretrained E0 encoder and 75-dimensional input interface are retained. Each removed feature dimension is replaced by its mean over valid labelled training epochs in that fold; the same replacement vector is applied to training, validation and test. TCN models are retrained for C, CP and CN with the specified E1 recipe and seed 42, while full CPN reuses E1. Thus the intervention changes the information available during both fitting and inference; it is not an inference-only deletion. The local protocol specifies the primary CPN–C boundary contrast and two secondary contrasts, with Holm correction across those three subject-wise Wilcoxon tests. Bootstrap intervals describe paired differences in pooled boundary macro-F1 under 10,000 subject-cluster resamples (seed 2028).

Table S17: Context-group ablation within E1 (seed 42). C, P and N denote current, previous and next epoch feature groups. Differences are calculated before rounding.

Condition	Overall macro-F1	Δ vs C	Boundary macro-F1	Δ vs C	N1 F1
C	.777477	—	.619102	—	.5011
CP	.777733	+.000257	.617686	-.001416	.5022
CN	.781585	+.004108	.618895	-.000207	.5055
CPN	.780230	+.002753	.620055	+.000953	.5115

For the protocol-specified CPN–C boundary comparison, Δmacro-F1 = 0.000953, 95% CI [−0.004588, 0.006568] and Holm p = 1.000 (36/78 subjects improved). Secondary CPN–CP and CPN–CN intervals also include zero with Holm p = 1.000. The result does not show that temporal context or P/N information is useless; it only finds no detectable marginal benefit for this encoding in E1.



(a) Silhouette by fold.

(b) Context-ablation effects.

Figure S1: Supporting representation and context analyses. (a) Silhouette coefficients by test fold. (b) Paired 95% confidence intervals for pooled boundary macro-F1 differences: CPN minus C, CP and CN (Sleep-EDF, E1, seed 42). All intervals include zero; Holm-adjusted Wilcoxon p = 1.000 for each contrast. Plots were drawn from archived aggregates using repository code with OpenAI Codex assistance.

Provenance

The historical SHHS protocol matches the original run-manifest hash (‘165d7cdf...fe93‘), not the expanded post-run audit (‘9541e233...fe9‘). Table S16 reuses seed-123 diagnostics matched to the EDF campaign and 20 E0/E3 fold-prediction files, and the SHHS1 extension manifest and 360 E0/E3 ensemble files. Hashes establish file identity, not independent replication or public preregistration. SHHS data and predictions remain externally held; E6 reanalyses locked predictions. Snapshot reconciliation is documented in `Reports/SHHS_PROTOCOL_PROVENANCE.md`.